fig01_state_decomposition.png

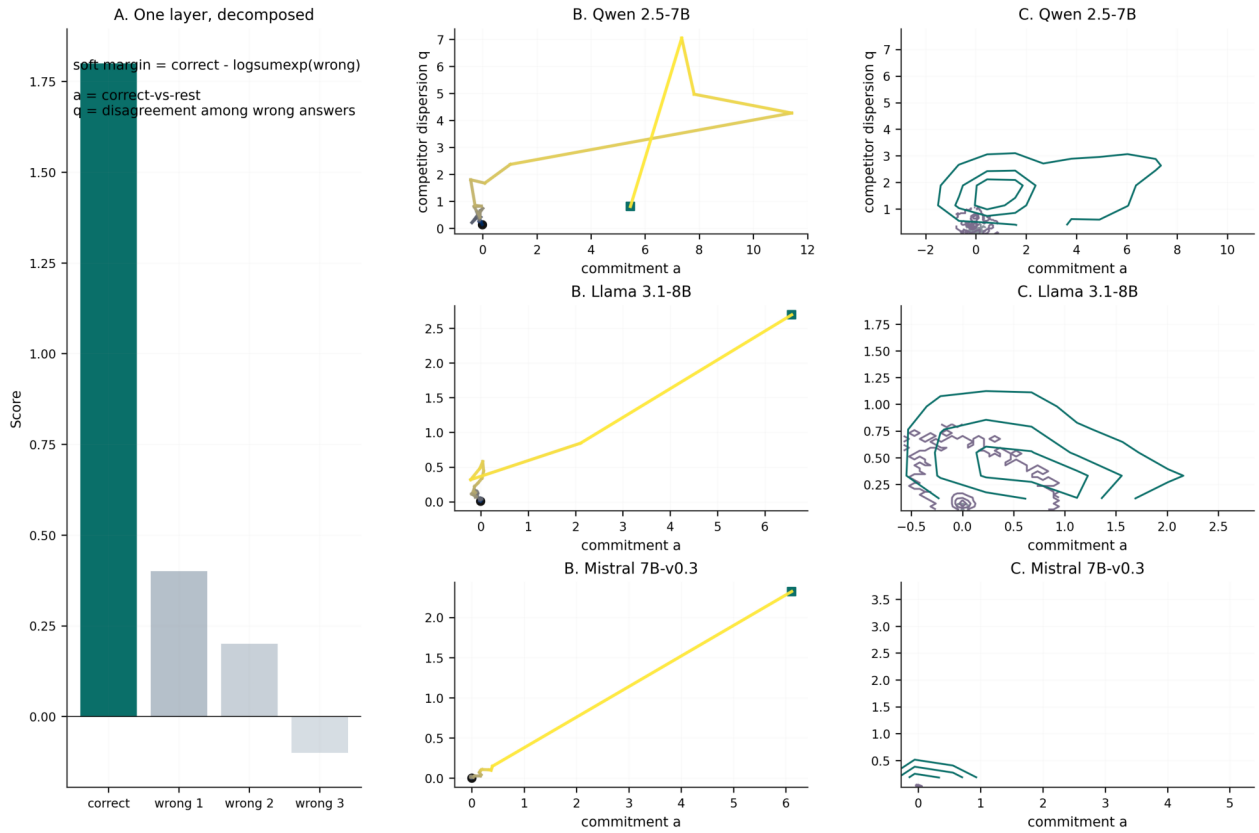


fig02_depthwise_dynamics.png

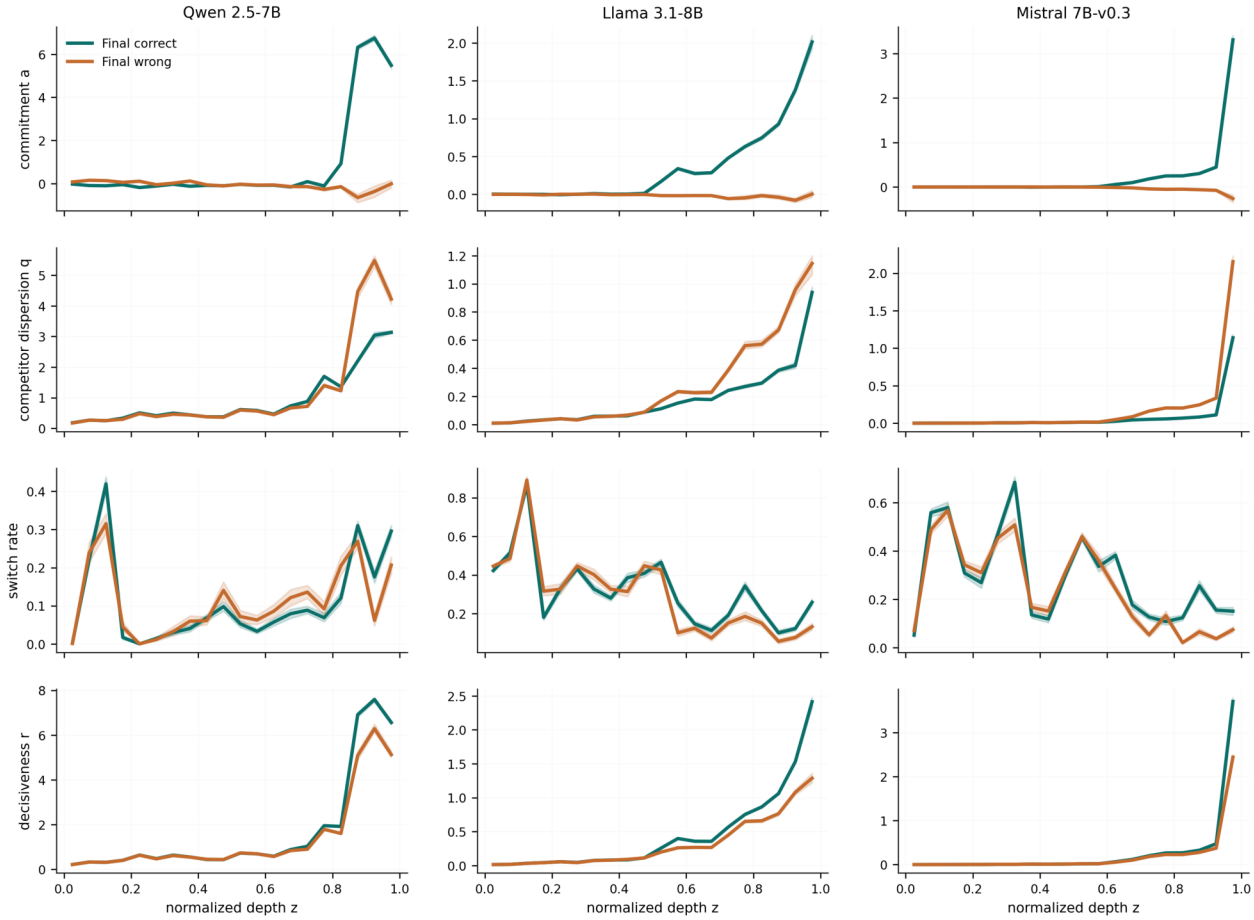


fig03_reversibility_map.png

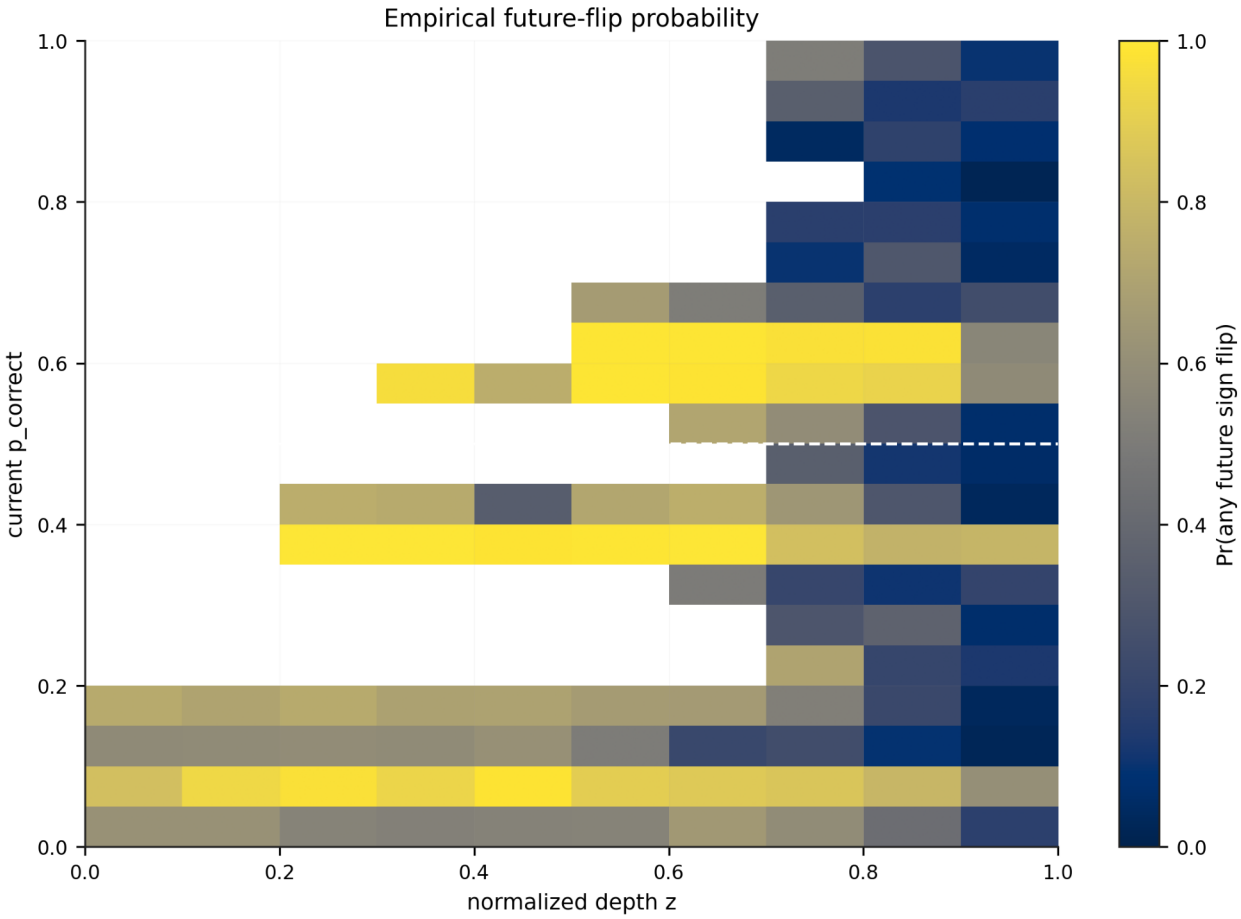


fig04_boundary_dwellers.png

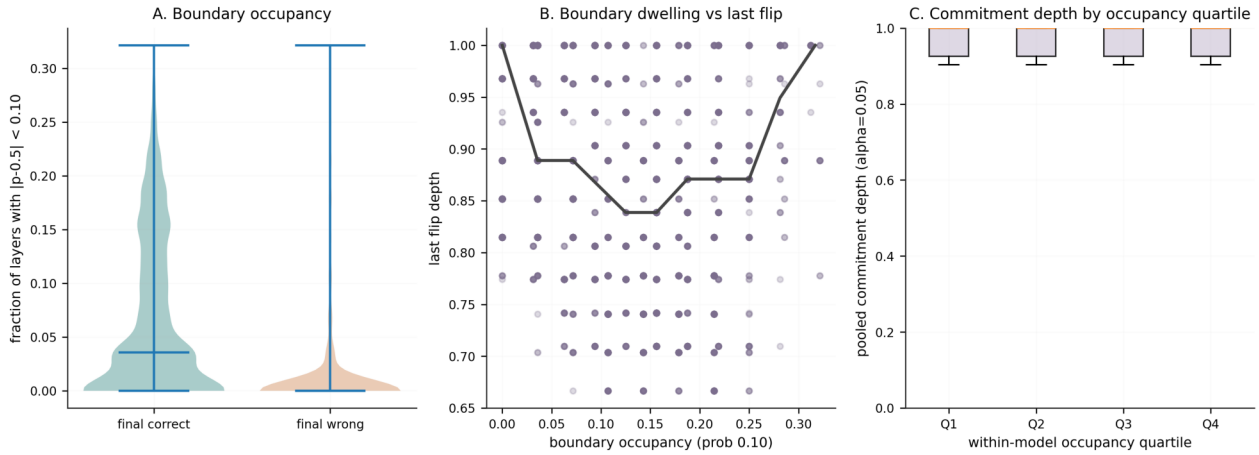
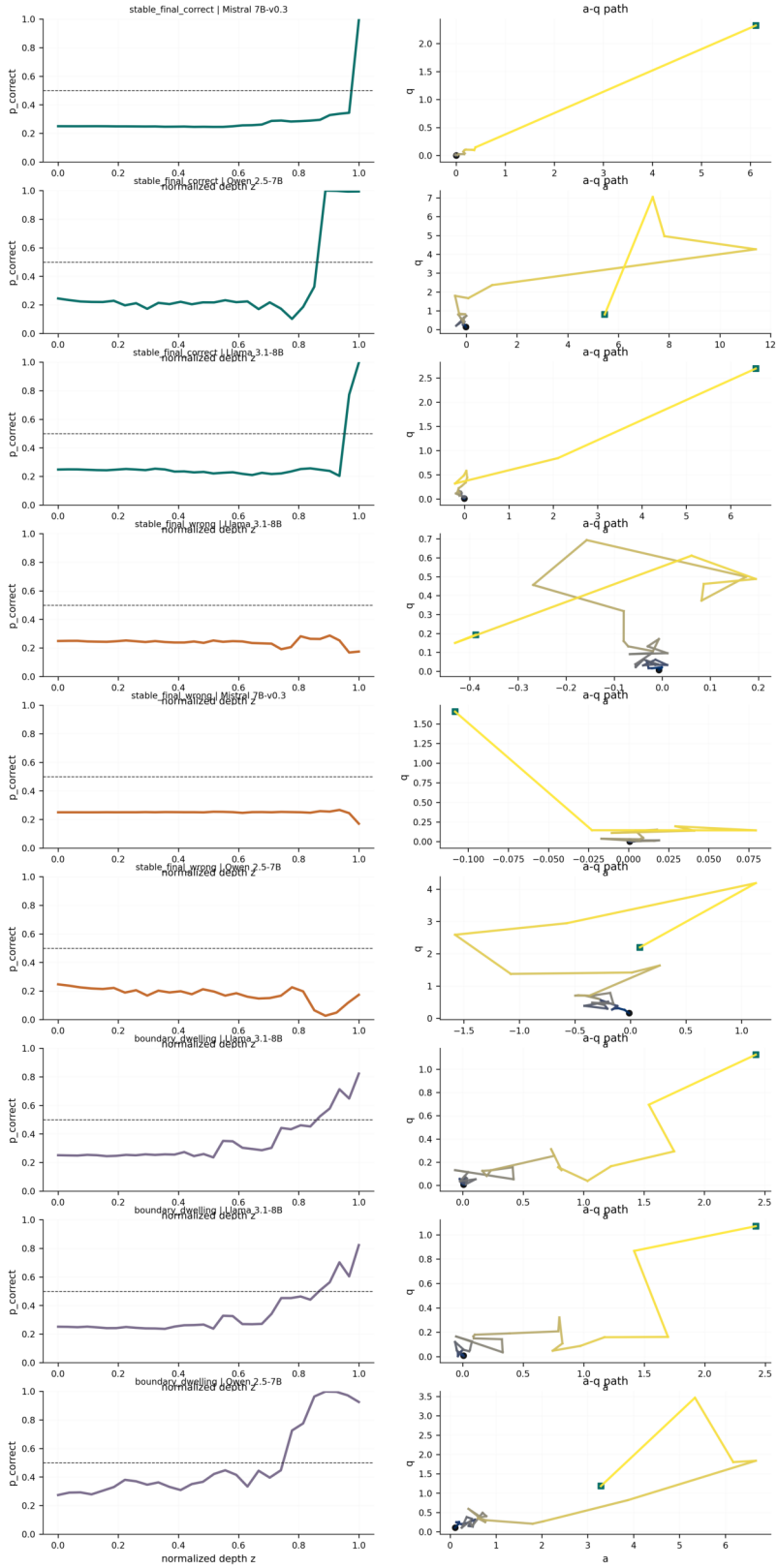
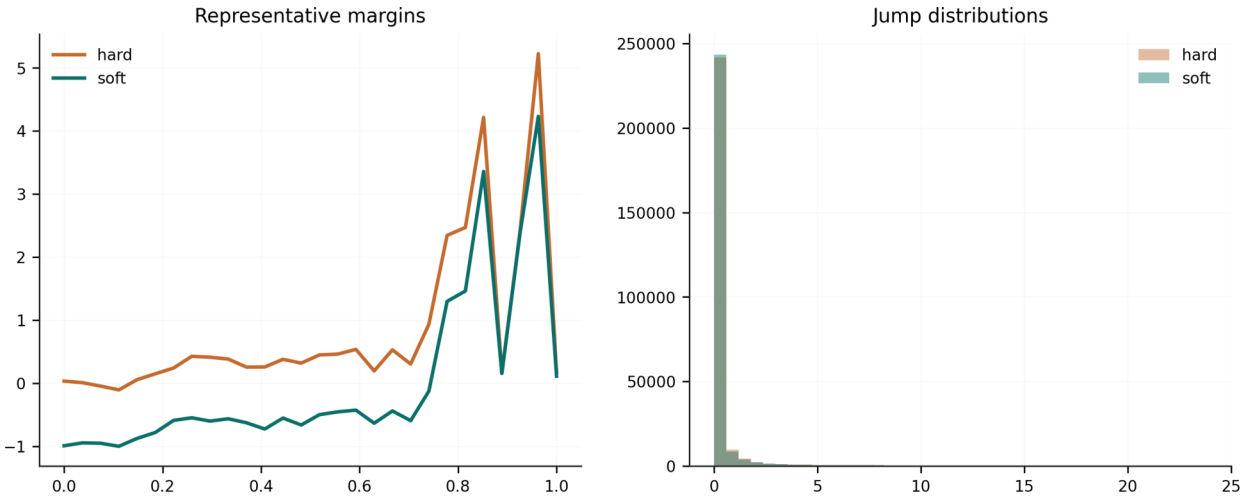


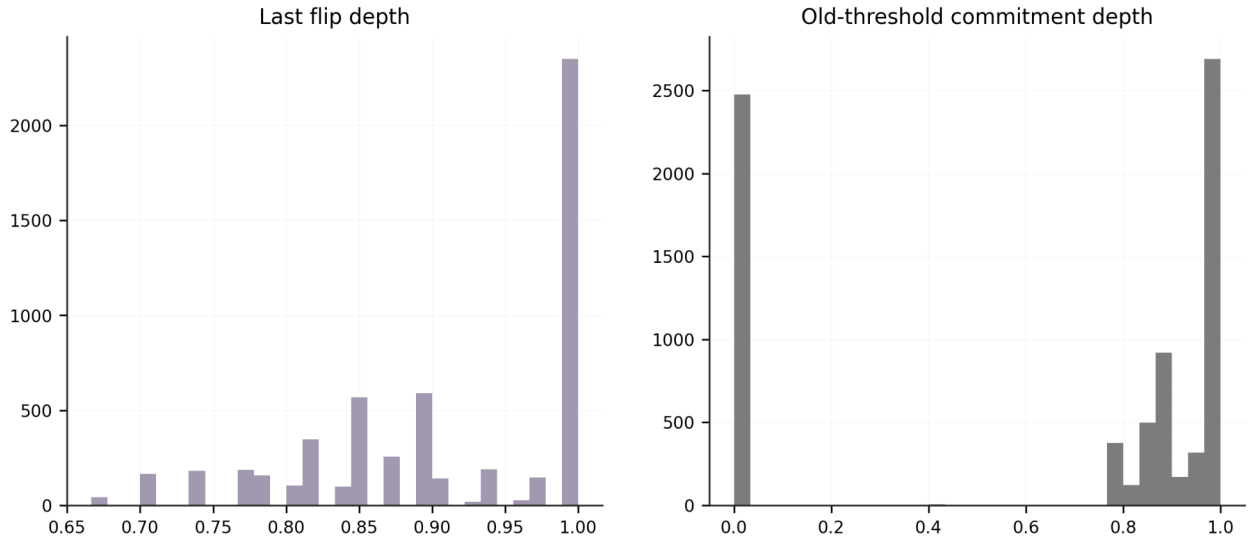
fig05_exemplar_trajectories.png



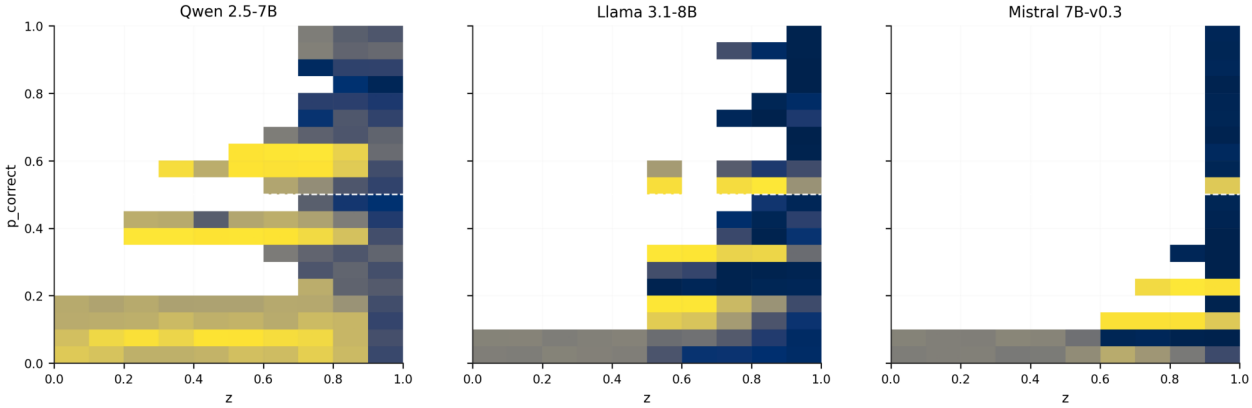
appendix_baseline_margin_smoothing.png



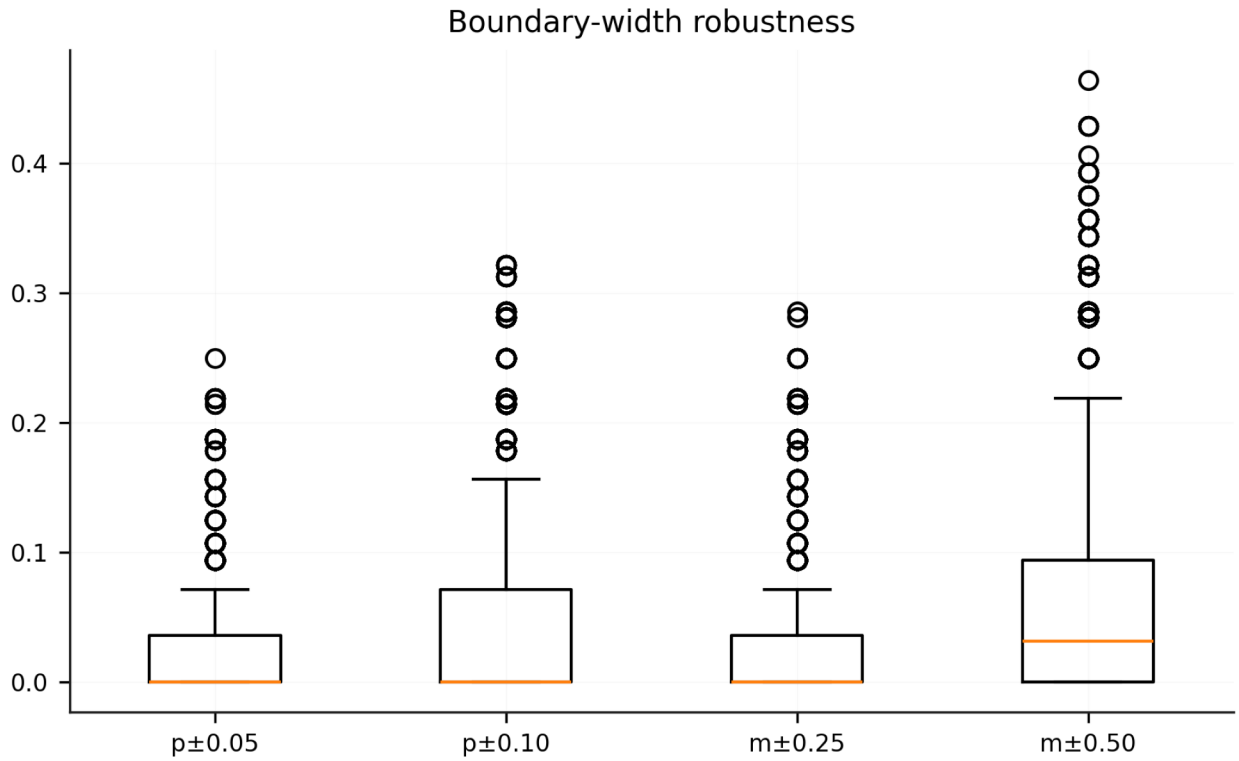
appendix_baseline_switch_commitment.png



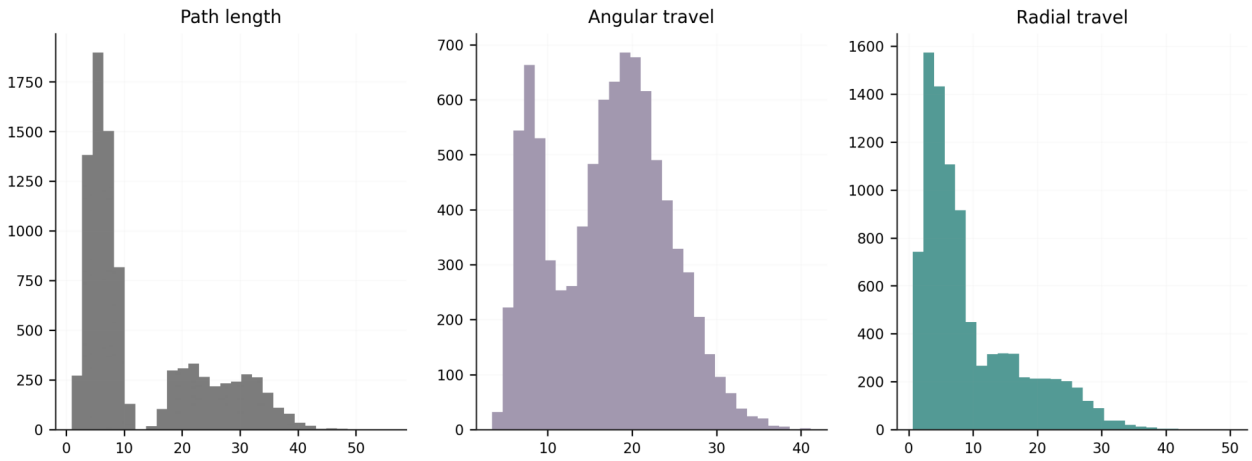
appendix_per_model_reversibility_maps.png



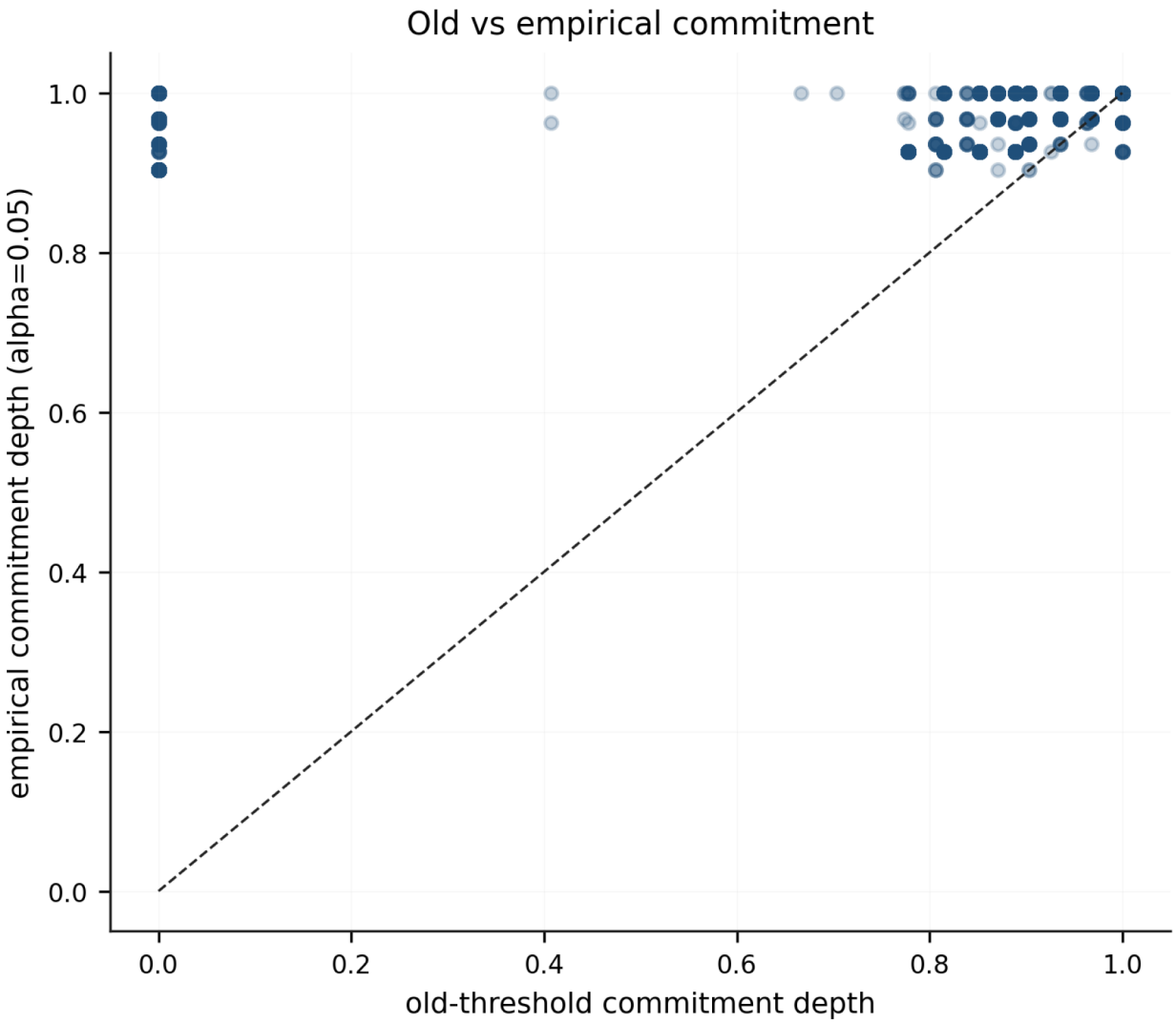
appendix_boundary_width_robustness.png



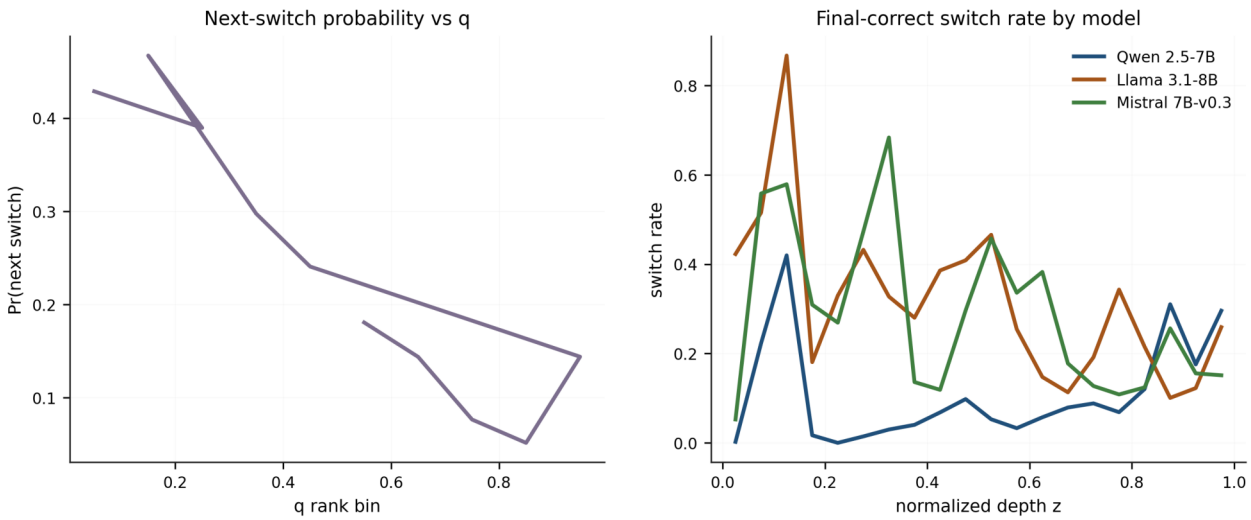
appendix_travel_metric_distributions.png



appendix_old_vs_empirical_commitment.png



appendix_competitor_dynamics.png



table_baseline_reproduction.csv

```csv
observation,scope,value,reproduced,details
soft_margin_smoother_than_hard_margin_at_switch_layers,pooled,0.002886365282008363,True,positive means hard jumps exceed soft
switch_rate_mid_or_earlier_peak_and_late_decline,pooled,0.39093333333333335,True,peak switch rate minus late-depth switch rate
old_regime_unstable_last_flips_are_late,pooled,1.0,True,median old last-flip depth in old unstable group
old_threshold_commitment_is_late,pooled,0.8888888888888888,True,median old-threshold commitment depth
```

table_boundary_correlations.csv

```
scope,x_metric,y_metric,n,spearman_rho,missing_y_count
pooled,boundary_occupancy_prob_010,last_flip_depth,5569,-0.4820138241864005,3431
pooled,boundary_occupancy_prob_010,flip_count,9000,0.5546263477635553,0
pooled,boundary_occupancy_prob_010,pooled_empirical_commitment_depth_005,4133,-0.029237256354189632,4867
Qwen/Qwen2.5-7B-Instruct,boundary_occupancy_prob_010,last_flip_depth,2421,-0.37901288390024895,579
Qwen/Qwen2.5-7B-Instruct,boundary_occupancy_prob_010,flip_count,3000,0.4427452725326944,0
Qwen/Qwen2.5-7B-Instruct,boundary_occupancy_prob_010,pooled_empirical_commitment_depth_005,1543,-0.23754646863227039,1457
meta-llama/Llama-3.1-8B-Instruct,boundary_occupancy_prob_010,last_flip_depth,1582,-0.11946523750347908,1418
meta-llama/Llama-3.1-8B-Instruct,boundary_occupancy_prob_010,flip_count,3000,0.8017809355473858,0
meta-llama/Llama-3.1-8B-Instruct,boundary_occupancy_prob_010,pooled_empirical_commitment_depth_005,1163,0.6405837245486329,183
mistralai/Mistral-7B-Instruct-v0.3,boundary_occupancy_prob_010,last_flip_depth,1566,,1434
mistralai/Mistral-7B-Instruct-v0.3,boundary_occupancy_prob_010,flip_count,3000,0.22019360253556153,0
mistralai/Mistral-7B-Instruct-v0.3,boundary_occupancy_prob_010,pooled_empirical_commitment_depth_005,1427,0.12313548031794251,
```

table_data_validation.csv

```
scope,metric,actual,expected,pass,details
global,canonical_rows,276000.0,276000.0,1,
global,manifest_rows,3000.0,3000.0,1,
global,manifest_prompt_uid_duplicates,0.0,0.0,1,
global,canonical_duplicate_rows,0.0,0.0,1,
global,missing_values_total,0.0,0.0,1,question may be empty string but not NA
Qwen/Qwen2.5-7B-Instruct,prompt_count,3000.0,3000.0,1,
Qwen/Qwen2.5-7B-Instruct,row_count,84000.0,84000.0,1,
Qwen/Qwen2.5-7B-Instruct,min_layer_index,0.0,0.0,1,
Qwen/Qwen2.5-7B-Instruct,max_layer_index,27.0,27.0,1,
meta-llama/Llama-3.1-8B-Instruct,prompt_count,3000.0,3000.0,1,
meta-llama/Llama-3.1-8B-Instruct,row_count,96000.0,96000.0,1,
meta-llama/Llama-3.1-8B-Instruct,min_layer_index,0.0,0.0,1,
meta-llama/Llama-3.1-8B-Instruct,max_layer_index,31.0,31.0,1,
mistralai/Mistral-7B-Instruct-v0.3,prompt_count,3000.0,3000.0,1,
mistralai/Mistral-7B-Instruct-v0.3,row_count,96000.0,96000.0,1,
mistralai/Mistral-7B-Instruct-v0.3,min_layer_index,0.0,0.0,1,
mistralai/Mistral-7B-Instruct-v0.3,max_layer_index,31.0,31.0,1,
global,complete_layer_coverage,1.0,1.0,1,
global,probability_sum_max_abs_error,4.440892098500626e-16,0.0,1,
global,decision_metrics_correct_key_mismatch,0.0,0.0,1,
global,decision_metrics_p_correct_max_abs_error,4.440892098500626e-16,0.0,1,
global,final_predicted_option_defined,1.0,1.0,1,
global,top_tied_rows,117.0,0.0,1,"reported for sensitivity, not a failure"
global,competitor_tied_rows,96.0,0.0,1,"reported for sensitivity, not a failure"
global,final_argmax_tied_trajectories,18.0,0.0,1,"reported for sensitivity, not a failure"
global,state_reconstruction_max_abs_error,1.0658141036401503e-14,0.0,1,
global,state_reconstruction_mean_abs_error,1.0843158192700291e-16,0.0,1,
global,p_correct_sigmoid_delta_soft_max_abs_error,5.551115123125783e-16,0.0,1,
global,q_below_theta_nan_threshold_count,0.0,0.0,1,
global,q_below_angular_threshold_count,0.0,0.0,1,
```

table_empirical_commitment_summary.csv

```
scope,final_group,alpha,pooled_nonmissing,pooled_never_enter,pooled_median_depth,model_nonmissing,model_never_enter,model_medi
pooled,all,0.05,4133,4867,1.0,5673,3327,1.0,7583,0.8888888888888888
pooled,final_correct,0.05,3150,2351,1.0,3919,1582,1.0,4482,1.0
pooled,final_wrong,0.05,983,2516,0.967741935483871,1754,1745,0.7741935483870968,3101,0.0
Qwen/Qwen2.5-7B-Instruct,all,0.05,1543,1457,0.9259259259259259,1230,1770,0.9259259259259259,2679,0.8518518518518519
Qwen/Qwen2.5-7B-Instruct,final_correct,0.05,1323,687,0.9259259259259259,1230,780,0.9259259259259259,1808,0.8888888888888888
Qwen/Qwen2.5-7B-Instruct,final_wrong,0.05,220,770,1.0,0,990,,871,0.0
meta-llama/Llama-3.1-8B-Instruct,all,0.05,1163,1837,1.0,1653,1347,1.0,2234,0.9354838709677419
meta-llama/Llama-3.1-8B-Instruct,final_correct,0.05,678,1170,1.0,1091,757,1.0,1249,1.0
meta-llama/Llama-3.1-8B-Instruct,final_wrong,0.05,485,667,0.9354838709677419,562,590,0.9032258064516129,985,0.0
mistralai/Mistral-7B-Instruct-v0.3,all,0.05,1427,1573,1.0,2790,210,1.0,2670,1.0
mistralai/Mistral-7B-Instruct-v0.3,final_correct,0.05,1149,494,1.0,1598,45,1.0,1425,1.0
mistralai/Mistral-7B-Instruct-v0.3,final_wrong,0.05,278,1079,1.0,1192,165,0.6129032258064516,1245,0.0
pooled,all,0.1,7076,1924,0.9259259259259259,6574,2426,0.9354838709677419,7583,0.8888888888888888
pooled,final_correct,0.1,4373,1128,1.0,4024,1477,1.0,4482,1.0
pooled,final_wrong,0.1,2703,796,0.9259259259259259,2550,949,0.8064516129032258,3101,0.0
Qwen/Qwen2.5-7B-Instruct,all,0.1,2568,432,0.8888888888888888,1660,1340,0.8888888888888888,2679,0.8518518518518519
Qwen/Qwen2.5-7B-Instruct,final_correct,0.1,1791,219,0.8888888888888888,1230,780,0.8888888888888888,1808,0.8888888888888888
Qwen/Qwen2.5-7B-Instruct,final_wrong,0.1,777,213,0.9259259259259259,430,560,0.9259259259259259,871,0.0
meta-llama/Llama-3.1-8B-Instruct,all,0.1,1931,1069,0.9354838709677419,2124,876,0.9354838709677419,2234,0.9354838709677419
meta-llama/Llama-3.1-8B-Instruct,final_correct,0.1,1184,664,1.0,1196,652,1.0,1249,1.0
meta-llama/Llama-3.1-8B-Instruct,final_wrong,0.1,747,405,0.9032258064516129,928,224,0.8064516129032258,985,0.0
mistralai/Mistral-7B-Instruct-v0.3,all,0.1,2577,423,1.0,2790,210,1.0,2670,1.0
mistralai/Mistral-7B-Instruct-v0.3,final_correct,0.1,1398,245,1.0,1598,45,1.0,1425,1.0
mistralai/Mistral-7B-Instruct-v0.3,final_wrong,0.1,1179,178,0.8064516129032258,1192,165,0.6129032258064516,1245,0.0
```

table_exemplar_prompts.csv

```
model_id,prompt_uid,subject,coarse_domain,question,correct_option,final_predicted_option,final_argmax_tie,final_correct,final_
mistralai/Mistral-7B-Instruct-v0.3,main::plain_exam::mmlu::test::1007,college_computer_science,computer_science,"If p(x) is th
Qwen/Qwen2.5-7B-Instruct,main::plain_exam::mmlu::test::1232,college_medicine,medicine,"An individual is born with a mutation c
meta-llama/Llama-3.1-8B-Instruct,main::plain_exam::mmlu::test::12169,professional_medicine,medicine,"A 25-year-old woman comes
meta-llama/Llama-3.1-8B-Instruct,main::plain_exam::mmlu::test::10867,professional_law,law,"A suspected drug dealer was arreste
mistralai/Mistral-7B-Instruct-v0.3,main::plain_exam::mmlu::test::10215,prehistory,history_politics,"Up to 50,000 years ago, wh
Qwen/Qwen2.5-7B-Instruct,main::plain_exam::mmlu::test::10526,professional_accounting,economics_finance_business,"The covarianc
meta-llama/Llama-3.1-8B-Instruct,main::plain_exam::mmlu::test::3216,high_school_computer_science,computer_science,"A large Jav
meta-llama/Llama-3.1-8B-Instruct,main::plain_exam::mmlu::test::1507,computer_security,computer_science,A digital signature nee
Qwen/Qwen2.5-7B-Instruct,main::plain_exam::mmlu::test::13819,virology,biology,PreP is most effective for which group:,A,A,Fals
```